Fractal Capability Contract – Info Sheet v0.1

Purpose

The Capability Contract is the universal agreement that every Operator Shell follows. It enables Fractal to host native tools, third-party plug-ins, and external professional software through one consistent interface.

Core Principle

Galaxy Shells organize recursive spaces. Operator Shells expose capabilities. The Capability Contract describes those capabilities independently of implementation.

Core Contract

Every operator declares:
• Identity
• Version
• Inputs
• Outputs
• Configuration
• Actions
• State
• Permissions
• Execution Environment
• Provenance
• Failure Behaviour

Input / Output Philosophy

Operators do not require a universal internal data format. They communicate through typed ports. Initial broad types: Text, Number, Boolean, Table, File, Image, Audio, Video, Document, Graph, Model, Geometry, Stream, Reference, Any.

Permissions

Declare only what is required (least privilege): project files, network, code execution, external software, GPU, remote compute, publishing.

Execution Environments

Native Fractal, Embedded Web Component, Local Kernel (Python/R/Julia), External Desktop Application, Remote Service, HPC/Cloud Compute, WebAssembly.

Examples

Text Operator → Native editor/viewer.
Regression Operator → Python kernel.
CFD Operator → External solver adapter.
AI Operator → Local or remote model.

Design Principles

1. One contract, many implementations.
2. Native operators form the baseline.
3. Third-party operators extend the ecosystem.
4. Complexity does not change the architecture.
5. The shell acts as an adapter between Fractal and capabilities.

Current Working Definition

An Operator Shell is a capability provider that presents, edits, translates, transforms, generates, or retrieves information through a declared capability contract.